FORMULA 1 · 2010–2027

The Economics of Winning

Predicting F1 Team Revenue and Championship Success from Financial & Performance Data

TEAM
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Motivation

F1 is a $2B+ industry where financial power and on-track success feed each other — prize money rewards winning, and bigger budgets build faster cars. **We ask two data questions: how much revenue will each team earn next season, and who will win the constructors' title?**

Group Project · SRH University of Applied Sciences Heidelberg

Prof. Mehrdad Jalali

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Dataset Description

Formula 1 Racing Industry 2010–2026 · Kaggle (sergionefedov)

Source: Kaggle — Formula 1 Racing Industry (2010–2026)

Structure: 5 relational tables (finances, business, circuits, standings, race results)

Unit of analysis: team–season panel (10 teams × 17 seasons)

Two targets built: prize money ($M) → regression; constructors' champion (0/1) → classification

164 team-season rows	**25** modelling features
17 seasons (2010–26)	**2** ML tasks

Why this dataset?

Rich mix of financial + performance features, a clean time dimension, and it naturally supports both a regression and a classification task from one source.

3

Preprocessing Steps

Every decision documented · leakage checked before modelling

1

Missing values

104 empty cost-cap values were structural (no cap before 2021) — filled with 0 as a sentinel, not imputed.

2

Outliers

IQR check on all money columns; extreme values kept as genuine (top teams really do out-spend).

3

Merge

Left-join team finances with F1-wide business totals on season — 164 rows preserved, no duplication.

4

Encoding

team_name one-hot encoded (drop_first) → Alpine baseline; boolean flags cast to 0/1.

5

Leakage check

total_revenue = sponsorship + prize money → dropped from features to avoid target leakage.

6

Split & scaling

80/20 split (stratified for classification); StandardScaler inside the model pipeline.

4

Feature Engineering & Selection

Domain features created, then ranked by four methods

Engineered features

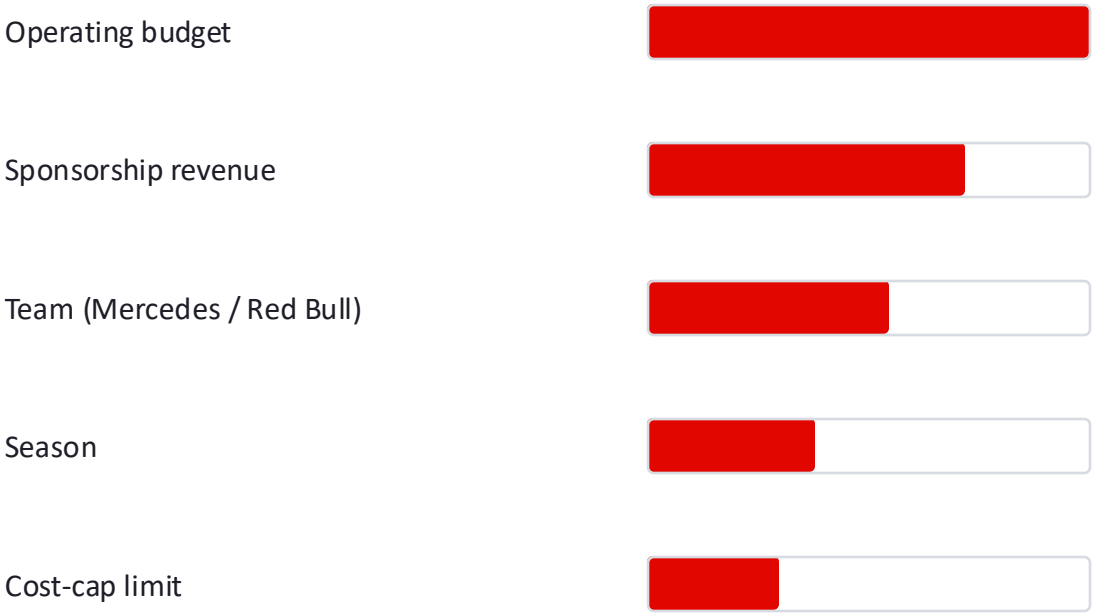
Sponsorship efficiency — *sponsorship ÷ budget*

Budget margin — *cost cap – budget*

Sponsorship % — *sponsorship ÷ budget × 100*

Cost-cap utilization — *budget ÷ cost cap*

Selected by correlation · mutual information · ANOVA · RF importance



Consistent across all four methods → budget dominates.

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Statistical Analysis And Data Visualization

Descriptive stats, normality, correlation & hypothesis testing

Methods applied

- Descriptive: mean, median, variance, std, skewness, kurtosis
- Normality: Shapiro–Wilk on all money columns (all non-normal)
- Correlation: Pearson & Spearman (budget ↔ prize money)
- Hypothesis tests: Welch t-test, ANOVA, Kruskal–Wallis, Chi-square
- confidence interval for mean prize money so instead of saying 'the average is 106 million,' I can honestly say 'we're 95% confident it's between about 98 and 115 million.'

6+ plots: histogram · boxplot · heatmap · violin · scatter · pairplot

r ≈ 0.9

Operating budget vs prize money (strong +)

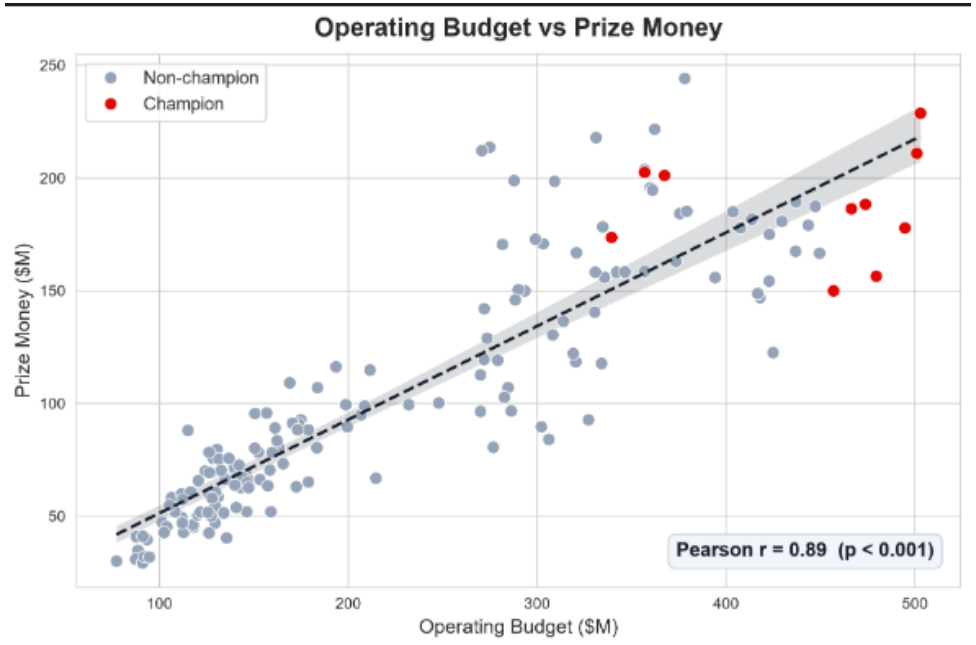
p < 0.05

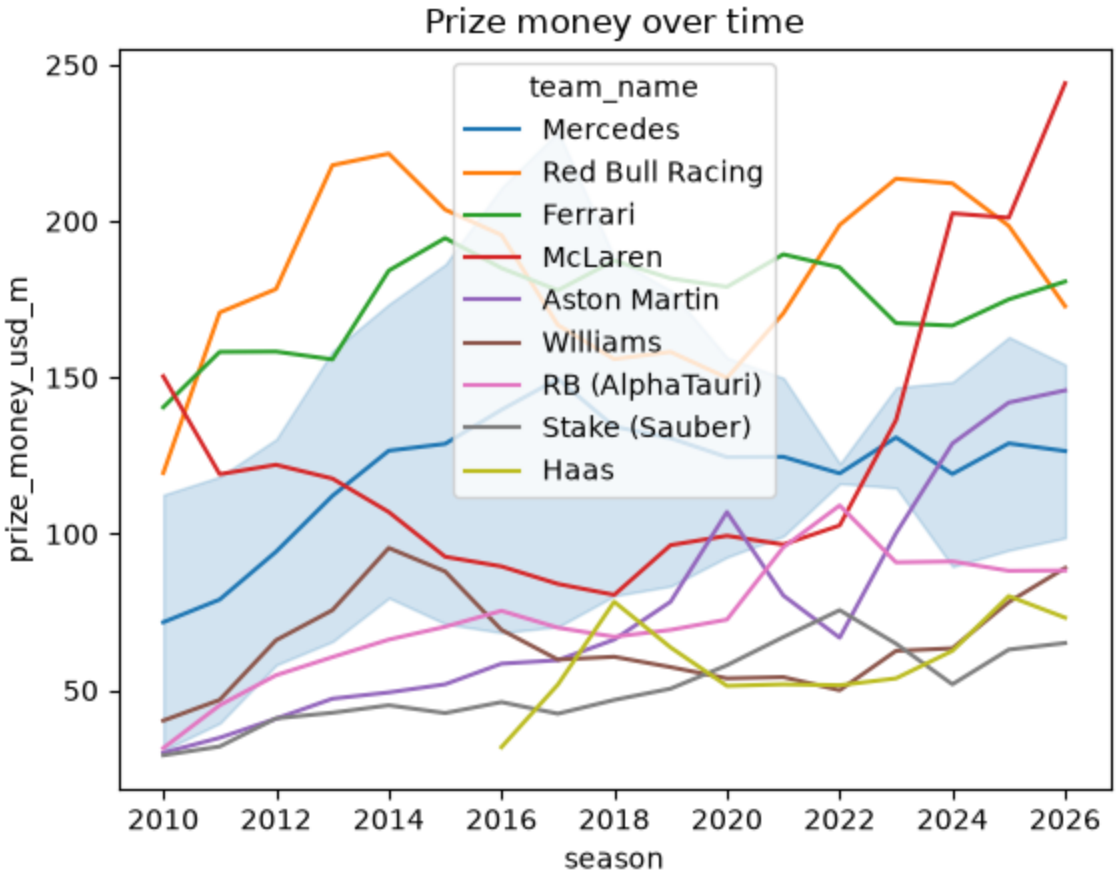
Champions out-spend the field (significant)

Non-normal

Money distributions are right-skewed

Data Visualization





6 Model Training:- Forecasting 2027 Team Revenue

A time-aware Lasso regression model for F1 Constructor revenue

What we are predicting

- Target: total_revenue_usd=sponsorship +prize money, per team per season
- Models: Naive, Linear, Ridge, Lasso, Random Forest
- Best Model: lasso
- Lag features (prior revenue & budget) + macro trends
- Split: train ≤2024, test 2025–26 (no random split)

Chosen: Lasso — best holdout accuracy

Model Selected

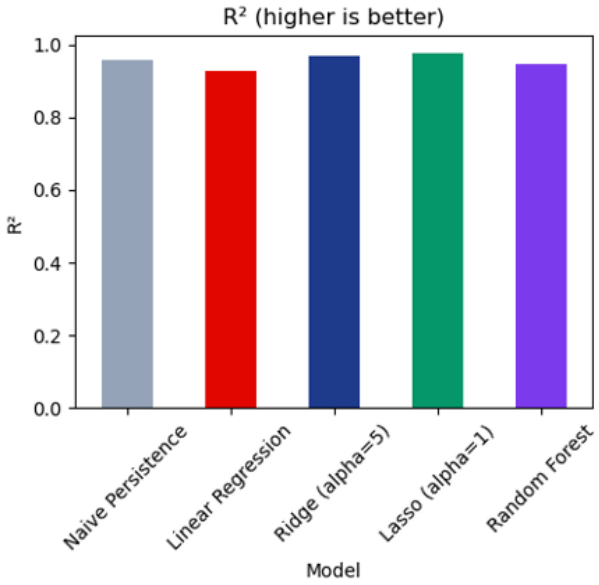
Lasso Regression($\alpha = 1$)

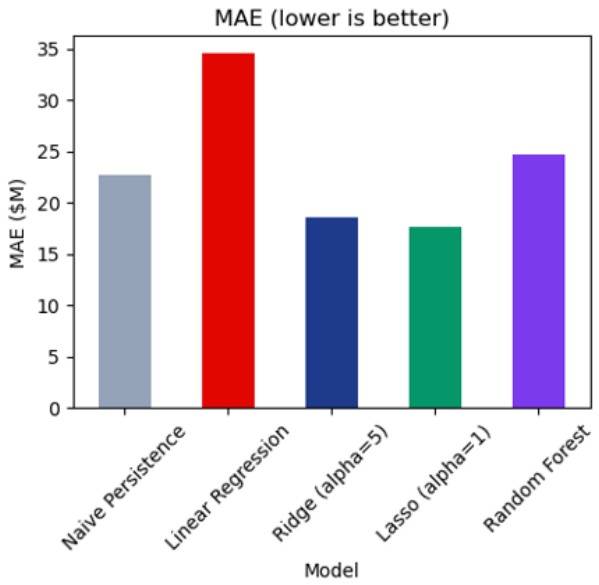
Chosen over liner Regression, ridge($\alpha = 5$), and Random Forest

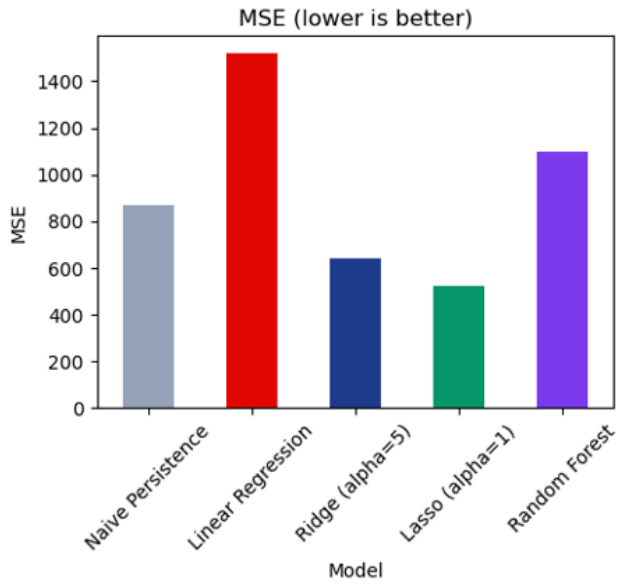
- Automatic feature selection: zeroed out every team dummy
- Outperformed Ridge across all 4 rolling validation windows (2023–2026), not just on average
- Kept only 8 of 19 features: lag1_revenue, operating budget, cost cap, season, and macro trend
- Refit on full 2011–2026 history for the final forecast (in-sample $R^2 \approx 0.98$)

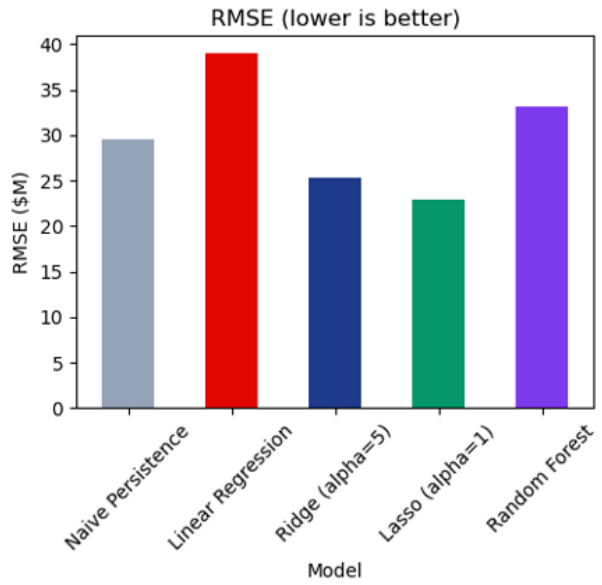
Model Evaluation

Held-out performance on 2025–2026 seasons, compared across four candidate models









Holdout R²

Best in class

Beats naive baseline

Yes — all windows

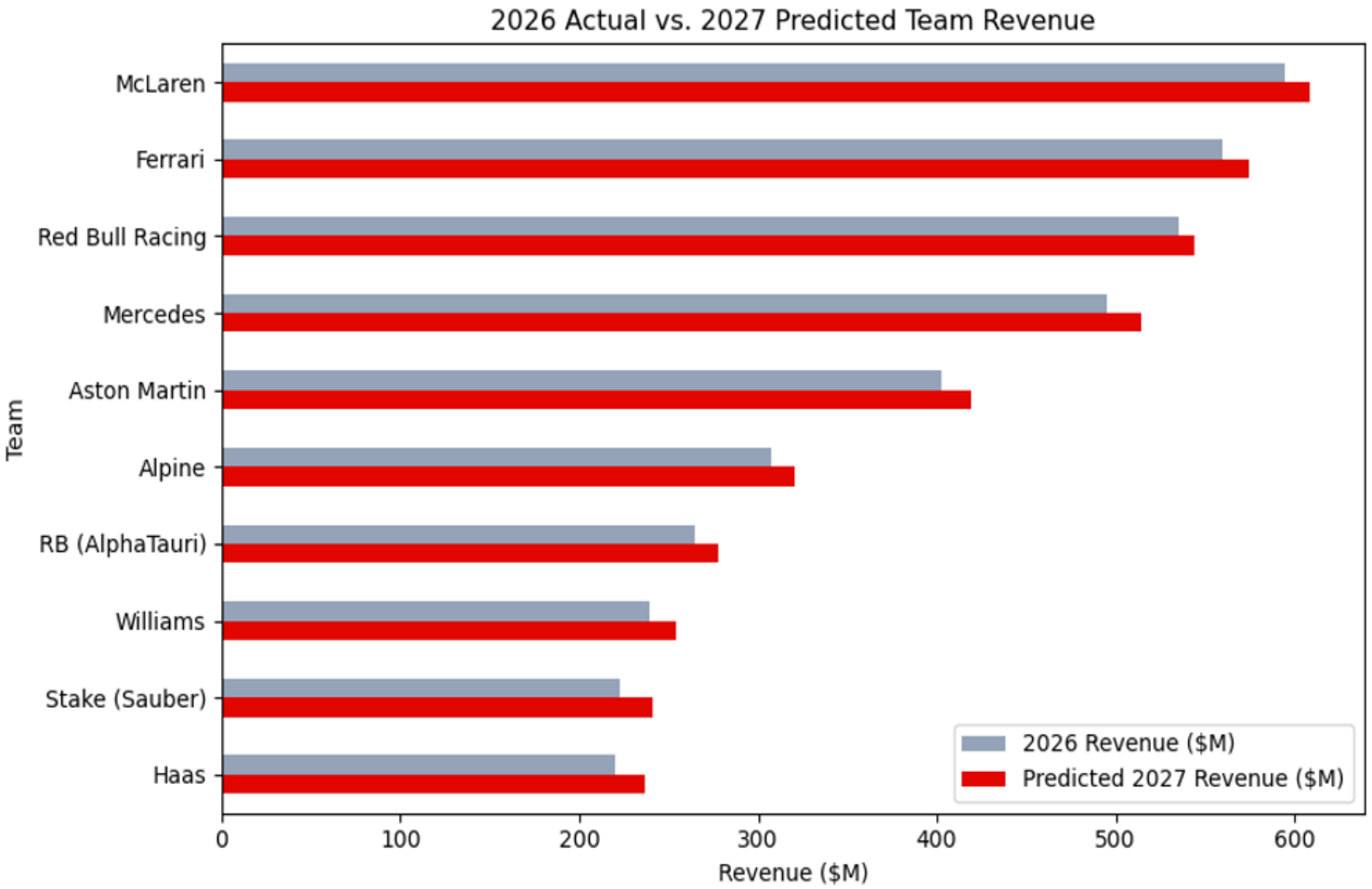
In-sample R²

≈ 0.98

Residual checks (actual vs. predicted, residuals vs. predicted, residual distribution) on the 2025–2026 holdout showed no systematic bias — errors were centered near zero with no strong pattern tied to prediction size.

2027 Revenue Forecast

Every constructor is projected to grow modestly year-over-year



Key Takeaways

- McLaren, Ferrari, Red Bull, and Mercedes stay the top 4 earners — no shift in the competitive financial hierarchy
- Revenue growth tracks last year's revenue/budget plus F1's overall commercial trajectory, not team brand identity
- This is a trend-based projection, not a guarantee — it assumes 2027 looks structurally like 2026 (same teams, same cost-cap regime)

Limitation: only 10 teams / 17 seasons (153 usable rows) — coefficients should be read directionally, not as precise dollar effects.

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Model Training

Two supervised tracks · time-based validation to prevent leakage

Classification — Champion

- Target: constructors' champion (0/1)
- Models: Logistic Regression, Random Forest, Naive
- Champion labels reconstructed from real history
- Rank teams by P(champion) → pick the top one

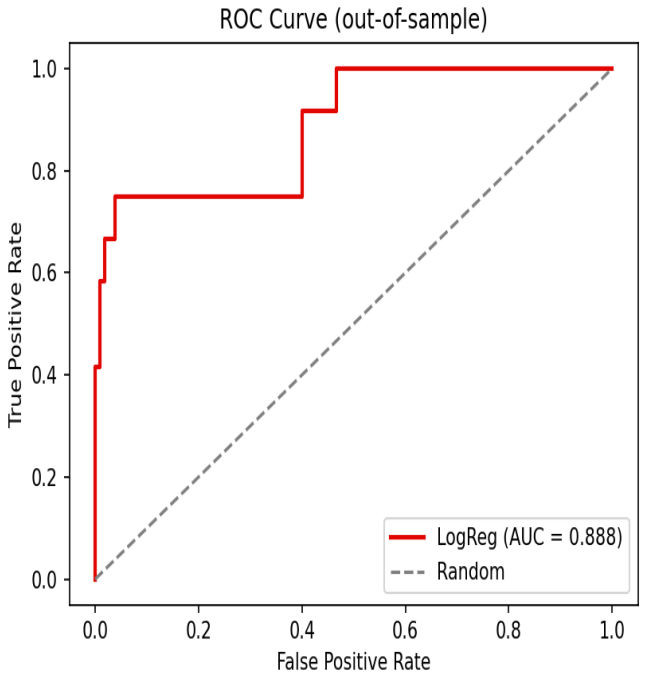
Chosen: Logistic Regression (interpretable)

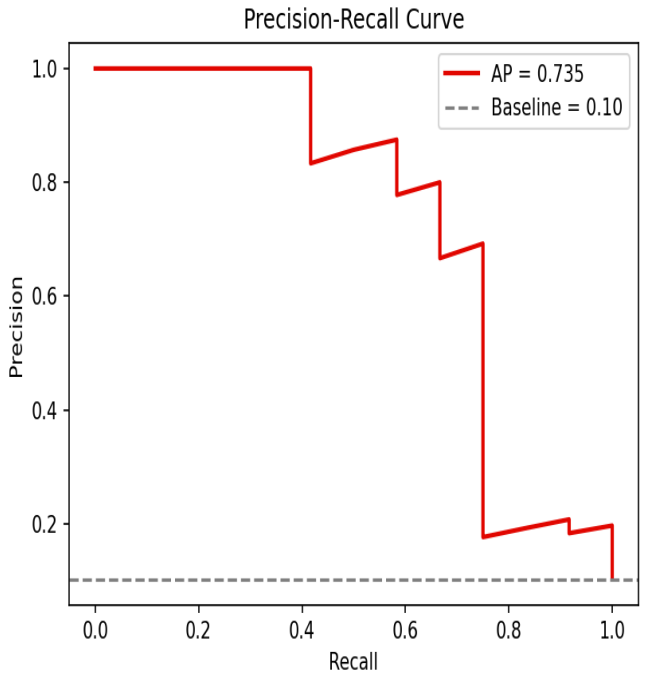
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Evaluation Results

Metrics that match each task — and honest baselines

Classification — champion pick





9 / 12 seasons' champion correct

0.91 ROC-AUC (Random Forest)

0.75 precision = recall = F1

Discussion & Interpretation

Key findings, model insights, and limitations of the ML approach applied to Formula One performance prediction

Key Insights

01 **Operating Budget**
Most influential predictor across both regression and classification models

02 **Financial–Performance Link**
Financial strength strongly correlates with on-track competitive performance

03 **Prior Season Performance**
Previous season results are highly predictive of future championship outcomes

04 **McLaren Identified**
Both models consistently identified McLaren as the strongest future championship candidate

05 **Model Generalisability**
ML effectively captures financial-performance dynamics in modern Formula One racing

Limitations

01 **Limited Champions Data**
Few championship-winning examples available, limiting classification training data

02 **Manual Label Reconstruction**
Historical championship labels were reconstructed manually, introducing potential inconsistencies

03 **Regulatory Assumptions**
Models assume stable F1 regulations; major rule changes may reduce predictive accuracy

04 **Future Uncertainty**
Predictions become less reliable the further into the future they are projected

05 **Dataset Scope**
Dataset spans 2010–2026 with 164 observations; broader historical data could improve robustness

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Roles & Contributions

Every member understands the full workflow; each owned one area

1

Boafo, Frederick

Dataset & Preprocessing

Data understanding, cleaning, merge, encoding, leakage checks

2

Shashank Kenkere Sridhar

Statistical Analysis

Descriptive stats, hypothesis tests, correlation & plots

3

Hrithik Wadile

Feature Engineering

Engineered features + selection (MI, ANOVA, importance)

4

Akshay Myakal,Sahil Suryawanshi

Modelling & Evaluation

Regression & classification training, metrics, validation

5

All

Interpretation & Report

Results synthesis, discussion, slides & report

Conclusion

Goal, model, key features, takeaways, and next steps

Goal

Use each team's F1 finances (2010–2026) to forecast 2027 revenue and predict the 2027 Constructors' Champion — turning historical budget, sponsorship, and prize-money data into a forward-looking prediction.

Best Model

Regression: Lasso (alpha=1) — beat Linear, Ridge, and Random Forest on every metric and every validation window.
Classification: Logistic Regression — tied Random Forest on ranking accuracy, chosen for interpretability.

Important Features

Operating budget was the #1 predictor across every method tested (correlation, mutual information, ANOVA, Random Forest). Sponsorship revenue, prior-season (lag) values, and season/macro trend followed. Team identity added little once these were known.

What We Learned

Money predicts money: budget and sponsorship explain 83% of prize-money variance (R2=0.833). But titles are 'sticky' — last year's champion beats financial features at predicting the next one, since our classifier could not beat the naive baseline.

Future Improvements

More seasons/teams to fix the small-sample and class-imbalance problems • add real-time factors (driver changes, rule changes, mid-season form) • un-chain the 2027 forecast instead of relying on a predicted 2026 champion • track live 2027 results to validate predictions as the season unfolds